STUDENT PERFORMANCE ANALYTICS

Final Capstone Project Report

Week 6 — Git, GitHub & Final Data Analytics Project

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1. Project Overview

This capstone project is submitted for the Week 6 assignment of the InternNova Data Analyst internship: **"Git, GitHub & Final Data Analytics Project."** It profiles a small student-performance dataset, validates and prepares the records, explores marks and attendance, produces labeled charts, and specifies a Power BI report. Python scripts (*data_prep.py* and *eda.py*) make the cleaning and analysis fully repeatable. The source records are synthetic and intended for practice, not for institutional decision-making.

The assignment objective is to apply the data analytics skills built across the internship — data cleaning, exploratory analysis, visualization, and Power BI dashboarding — in a single end-to-end project, while also practicing a basic Git and GitHub workflow and producing a project that can be added to a portfolio.

What this report covers

- Dataset description and the validation rules applied to it
- The data cleaning / preparation process and its results on this file
- Exploratory data analysis: descriptive statistics, correlation, and outlier screening
- Six labeled visualizations covering distribution, comparison, composition, and relationship views
- The Power BI dashboard build specification (KPIs, visuals, measures, slicers, layout)
- Evidence-based business insights and recommendations
- Conclusion and scope limitations

2. Problem Statement

Summarize achievement and attendance by subject and department, identify where additional academic support may be useful, and communicate the findings in a form that can be explored in Power BI. The supplied data is **cross-sectional** (a single snapshot, no date or term field), so the analysis cannot answer questions about change over time — only about differences across the students, subjects, and departments present in this one file.

3. Dataset Description

The raw file is *data/raw/students.csv*, a synthetic dataset carried over from an earlier student-performance project. It contains **24 student records** and **8 columns**. Marks and attendance are measured on a 0–100 scale. Four departments are represented equally, six students each: Engineering, Computer Science, Arts, and Science.

Column	Type	Description
student_id	Text	Unique student identifier (e.g. STU001)
name	Text	Student full name
department	Categorical	One of 4 departments — 6 students each
english / mathematics / science / computer_science	Numeric (0–100)	Subject mark

Column	Type	Description
attendance	Numeric (0–100)	Attendance percentage

The source file has **no missing values, duplicate rows, duplicate IDs, or date/term column**. Every identity field is populated and every mark and attendance value falls inside the valid 0–100 range.

4. Data Cleaning & Preparation Process

scripts/data_prep.py applies a documented, repeatable validation pipeline before any analysis is run:

- **Identity fields** (student_id, name, department): trim surrounding whitespace; remove any record with a blank or missing identity/category field, since it cannot be reliably identified or grouped.
- **Numeric fields** (4 subject marks, attendance): coerce to numeric; drop records with values outside the valid 0–100 range rather than silently correcting them.
- **Missing numeric values**: fill from the department median for that column, falling back to the cohort median if the department has no observed value.
- **Duplicates**: remove exact duplicate rows; if a student ID still repeats, keep the first valid record so the key stays unique.

Result on this file

Running the pipeline on the supplied data found **no issues to correct**: no missing values, no out-of-range measures, no duplicate rows, and no blank identity fields. All 24 records were retained without imputation, and *students_cleaned.csv* is identical to the raw input. The table below is the validation log that the script prints and that this run produced.

Validation check	Result
Exact duplicate rows removed	0
Rows with missing identity fields removed	0
Rows with out-of-range measures removed	0
Repeated student IDs removed (first valid kept)	0
Missing values imputed (per numeric column)	0 each
Final cleaned shape	24 rows × 8 columns

5. Exploratory Data Analysis

scripts/eda.py computes descriptive statistics, a correlation matrix, and IQR-based outlier screening on the cleaned data, and derives each student's average mark and pass/fail result (pass requires every subject mark to be at least 40).

5.1 Descriptive statistics

	Eng.	Math.	Sci.	C.Sci.	Attend.	Avg.
count	24.00	24.00	24.00	24.00	24.00	24.00
mean	70.67	70.83	72.54	74.83	85.83	72.22

	Eng.	Math.	Sci.	C.Sci.	Attend.	Avg.
std	18.07	17.86	15.85	16.27	8.53	15.87
min	39.00	39.00	40.00	40.00	68.00	40.00
25%	57.25	58.75	61.75	62.50	79.75	61.19
50%	73.50	72.50	73.50	74.00	86.50	73.75
75%	84.50	85.50	86.50	88.25	92.25	86.13
max	99.00	96.00	94.00	100.00	99.00	97.25

Eng. = English · Math. = Mathematics · Sci. = Science · C.Sci. = Computer Science · Attend. = Attendance (%) · Avg. = student average mark

5.2 Correlation matrix (Pearson r)

	Eng.	Math.	Sci.	C.Sci.	Attend.	Avg.
English	1.000	0.770	**0.904**	0.884	**0.920**	**0.954**
Mathematics	0.770	1.000	0.746	0.754	0.807	0.880
Science	**0.904**	0.746	1.000	**0.918**	**0.954**	**0.952**
Comp. Sci.	0.884	0.754	**0.918**	1.000	**0.933**	**0.949**
Attendance	**0.920**	0.807	**0.954**	**0.933**	1.000	**0.966**
Avg. Marks	**0.954**	0.880	**0.952**	**0.949**	**0.966**	1.000

Attendance correlates most strongly with average marks (**r = 0.966**) and with science (**r = 0.954**). This is a descriptive association in a 24-record synthetic cohort, not evidence that attendance *causes* higher marks.

5.3 Outlier screening (1.5 × IQR rule)

Every numeric column — each subject mark, attendance, and each student's average mark — was screened against its own 1.5×IQR fences. **Zero observations were flagged.** A zero count means no value crossed its fence; it does not prove every record is error-free, only that nothing in this sample is numerically extreme relative to its own column.

5.4 Key findings

- The cleaned dataset holds **24 students across 4 departments** with no date field, so this analysis makes no time-based trend claim.
- The cohort average is **72.22 / 100**; average attendance is **85.83%**.
- **21 of 24 students (87.5%)** pass under the rule that every subject mark must be at least 40; **3 fail**.
- **Engineering** has the highest department average (**78.46/100**, n=6, 100% pass); **Science** has the lowest (**65.79/100**, n=6) and a **50.0%** pass rate.
- The strongest subject by mean is **Computer Science (74.83)**; the weakest is **English (70.67)** — a 4.17-point gap.
- The Pearson correlation between attendance and student average marks is **0.966**: a strong association in this small synthetic cohort, not proof of causation.

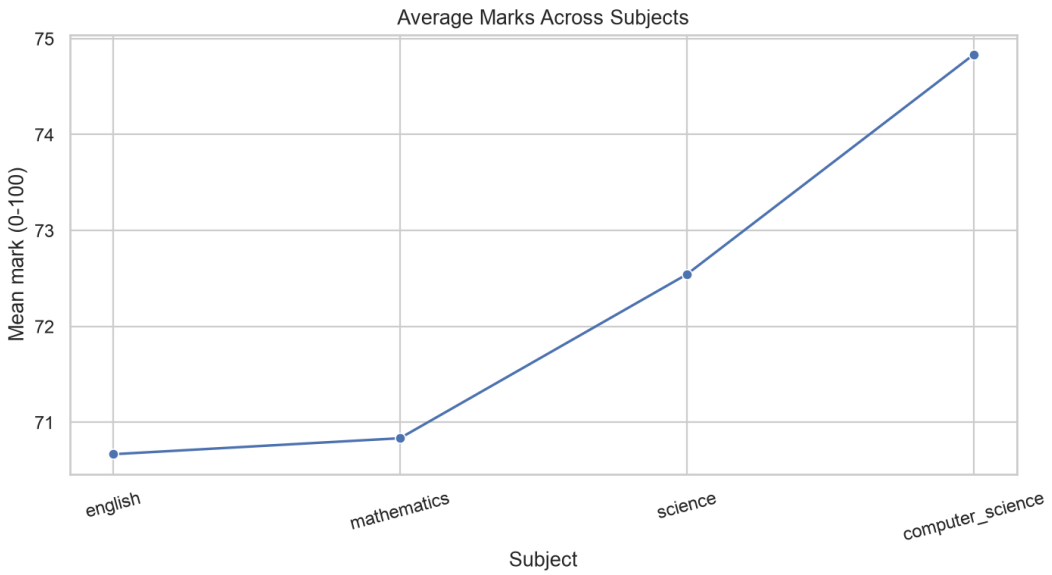
Interpretation limits: the supplied file is a small synthetic cross-sectional dataset. Department differences are descriptive, not causal or necessarily representative of a real school. With no date/term field, year-over-year or

within-student change cannot be measured.

6. Data Visualizations

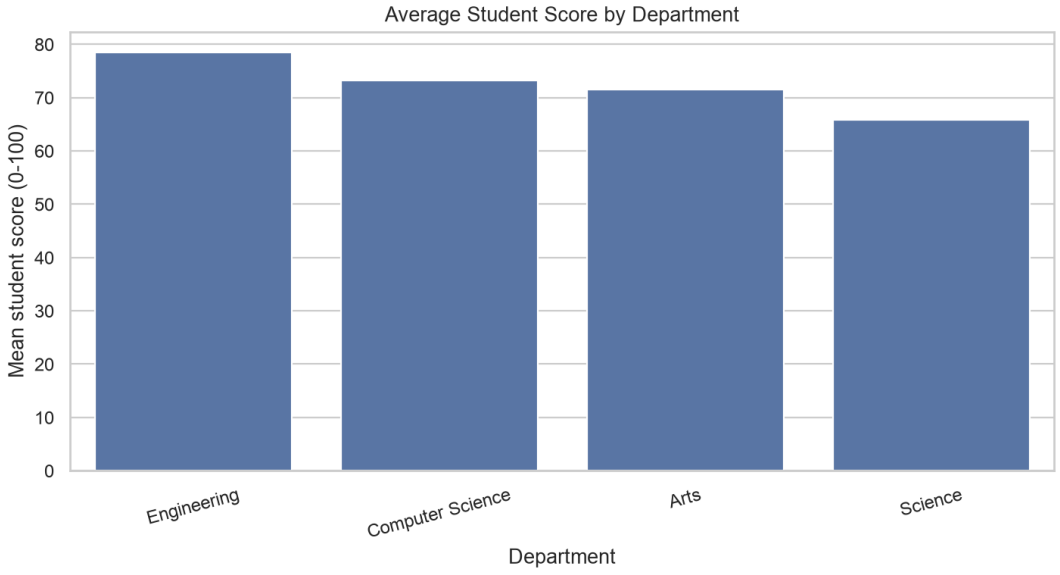
Six labeled charts were produced from the cleaned data, each with a descriptive title and axis labels, covering distribution, comparison, composition, spread, and relationship views.

Figure 1. Average Marks Across Subjects



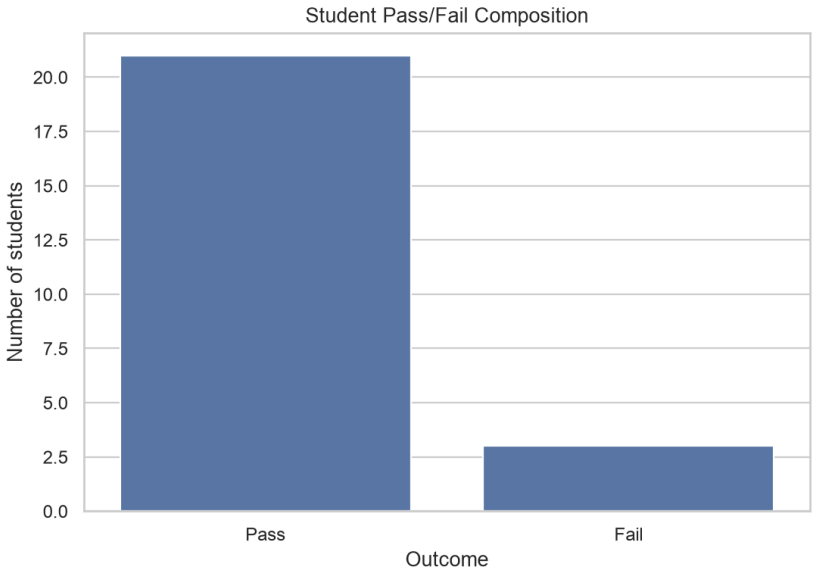
Computer Science has the highest subject mean (74.83); English the lowest (70.67). The connecting line compares categories — it is not a time series.

Figure 2. Average Student Score by Department



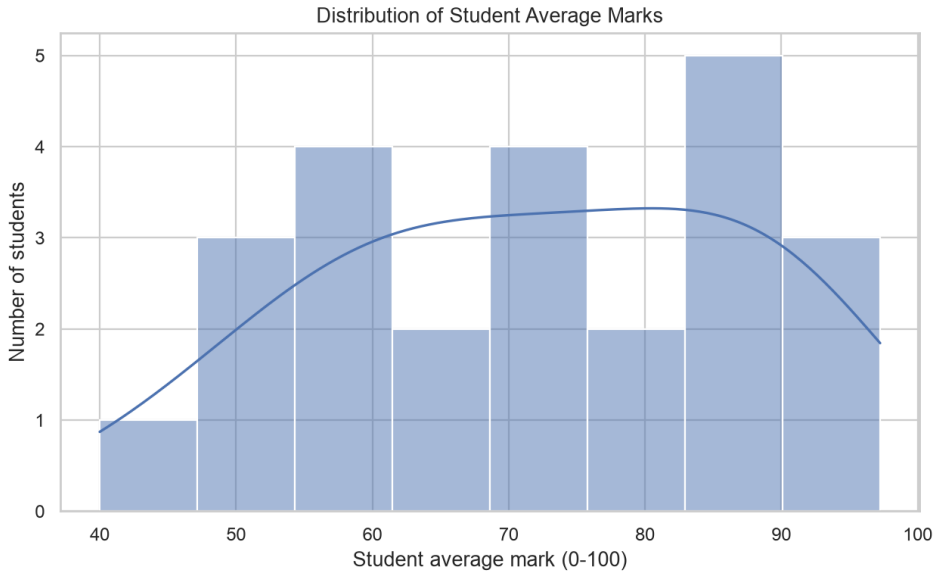
Engineering leads at 78.46; Science trails at 65.79, a 12.67-point gap between the strongest and weakest department.

Figure 3. Student Pass/Fail Composition



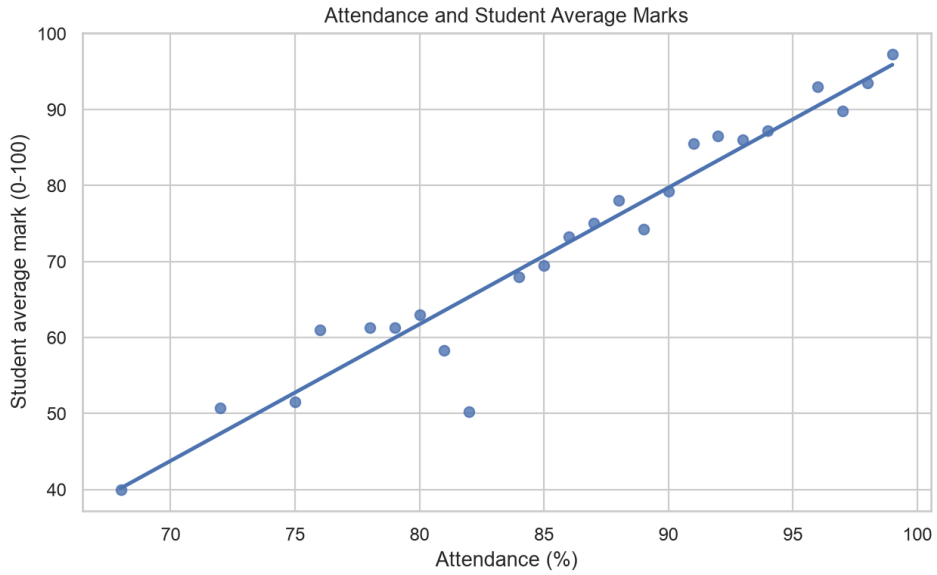
21 students pass and 3 fail under the all-subjects-≥40 rule — an 87.5% overall pass rate.

Figure 4. Distribution of Student Average Marks



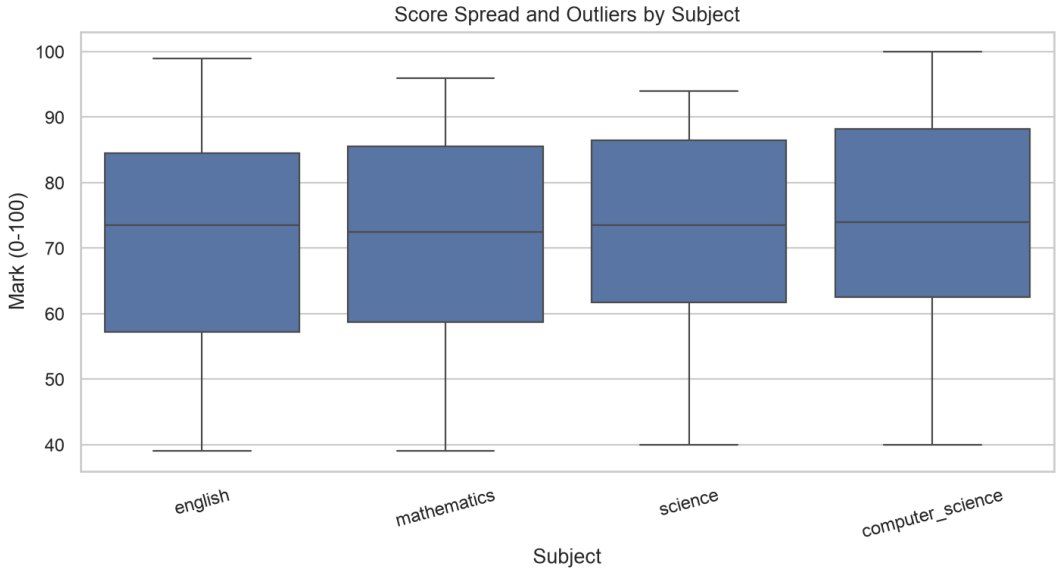
Student averages spread fairly evenly from 40 to 97, with a cluster of high performers in the 83–90 range.

Figure 5. Attendance and Student Average Marks



A clear upward relationship: Pearson r = 0.966. Association only — not a causal claim — given the small, synthetic, cross-sectional sample.

Figure 6. Score Spread and Outliers by Subject



Medians sit near 72–74 for all four subjects; the 1.5×IQR screen found no outlier points in any subject.

7. Power BI Dashboard

The section below is the **build specification** that drives the Power BI report: the Power Query transformation, the DAX measures, every visual, and the page layout, exactly as implemented in Power BI Desktop.

Add before final submission: *screenshots of the live Power BI Desktop report were not among the files supplied for this document, so none are inserted here. Export each report page (File → Export → PDF, or a screen capture) from the .pbix and paste the image(s) into this section before the PDF is submitted.*

7.1 Data source and transformation

Power Query loads the raw CSV, trims and validates identity fields, coerces and range-checks the numeric fields, fills any missing measure by department median (cohort median as fallback), removes duplicates, derives a pass/fail flag, and unpivots the four subject columns into long format as the *StudentScores* table. Because the supplied CSV is already complete and valid, the query output is **96 rows** (24 students × 4 subjects) with no imputation or row removal.

7.2 KPI cards (4)

- **Student Count** — DISTINCTCOUNT of Student ID
- **Average Student Score** — mean of each student's average mark
- **Pass Rate** — share of distinct students with PassFlag = 1
- **Average Attendance** — mean attendance across all rows

7.3 Supporting measures

- Average Subject Mark, Department Average Score, Department Pass Rate, Failing Students (distinct count where PassFlag = 0)

7.4 Visuals (5)

- **Subject comparison** — clustered column, Subject × Average Subject Mark
- **Department performance** — bar chart, Department × Department Average Score
- **Pass rate by department** — column chart, Department × Department Pass Rate (%)
- **Attendance vs. achievement** — scatter, Attendance (X) × Average Student Score (Y), Student ID in Details, Department in Legend
- **Top students** — table of Student Name, Department, Average Student Score, with a Top 10 visual-level filter

7.5 Slicers and layout

Slicers: **Department, Subject, Student Name, PassFlag** (labeled Pass/Fail). Single-page 16:9 report: four KPI cards across the top; department performance and subject comparison in the middle row; the attendance-vs-achievement scatter and Top 10 table in the lower row; slicers in a compact left rail. No time slicer or time-series visual is included — the source has no date field — and the report carries a note that the data is a 24-record synthetic sample.

8. Business Insights & Recommendations

8.1 Evidence-based insights

- **The overall cohort is above the minimum pass threshold, but not uniformly strong.** The mean student average is 72.22/100; 21 of 24 students (87.5%) pass when every subject must be at least 40. Three students (12.5%) fail.
- **Science is the clearest department-level support priority in this sample.** Its six students average 65.79, the lowest department mean, and only three of six pass (50%). Engineering averages 78.46 with all six students passing — a 12.67-point gap versus Science.
- **English is the lowest-mean subject; Computer Science is the highest.** English averages 70.67 and Computer Science 74.83, a 4.17-point difference. This suggests reviewing English outcomes first, though the gap alone does not diagnose a cause.
- **Attendance and marks move together strongly in this dataset.** The Pearson correlation between attendance and student average marks is 0.966 across 24 records — a descriptive association only; it does not establish that attendance causes higher marks, and the data is synthetic and small.
- **The IQR rule flags no numeric outliers.** No subject mark, attendance value, or student average falls outside its 1.5×IQR fences. This makes the present distributions easy to summarize, but does not rule out data-entry errors that remain inside valid ranges.
- **The file supports group comparisons, not longitudinal conclusions.** There is no date or term field, so the analysis cannot identify improvement or decline over time, seasonal effects, or within-student change.

8.2 Recommendations

- **Run a focused diagnostic for the Science department.** Review the three failing students' subject-level marks, identify the topics causing scores below 40, and provide targeted tutoring. Recheck the department pass rate after the next assessment before scaling the intervention.
- **Review English assessment results and add term/date plus student-level follow-up data for future reporting.** English has the lowest subject mean (70.67); use item-level evidence before changing curriculum or allocating resources, and track attendance alongside achievement without treating correlation as proof of cause.

Scope note: these are practice recommendations from 24 synthetic records, not claims about a real student population. Department labels may not describe comparable cohorts; consult educators and validate data before acting on decisions that affect students.

9. Conclusion

This project demonstrates a repeatable analysis workflow from raw CSV validation to documented findings, saved visualizations, and a Power BI build specification. The results are descriptive and intended for practice only. A real deployment should use a larger, quality-checked dataset with dates/terms, privacy controls, and stakeholder review.

Overall, the cohort passes comfortably on average (72.22/100, 87.5% pass rate), but the gap between Engineering and Science — and between Computer Science and English — points to where support is best targeted next. Attendance tracks closely with achievement in this sample and is worth monitoring, though not treating as a guaranteed lever, without further data.

Tools used

- Python 3 — pandas for CSV handling, validation, and grouping
- matplotlib and seaborn for the six labeled figures
- Power BI Desktop — Power Query (M) and DAX for the dashboard
- Git and GitHub for version control and submission

End of report — Student Performance Analytics, Week 6 Capstone, InternNova Data Analyst Internship.